



WATER APPLICATION VALUE ENGINE

WATER SOLUTIONS

WAVE Program Version: 1.85.29

Calculation Engine Version: 01.12.40.01

Database Version: 90



Project Name: Wave-UF_RO project
Case Name: Case 1
Customer:
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Company: DuPont
Country:
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Project Notes: Surface water treatment

Case #: 1 of: 1
Case Notes: Case 1

Keywords:

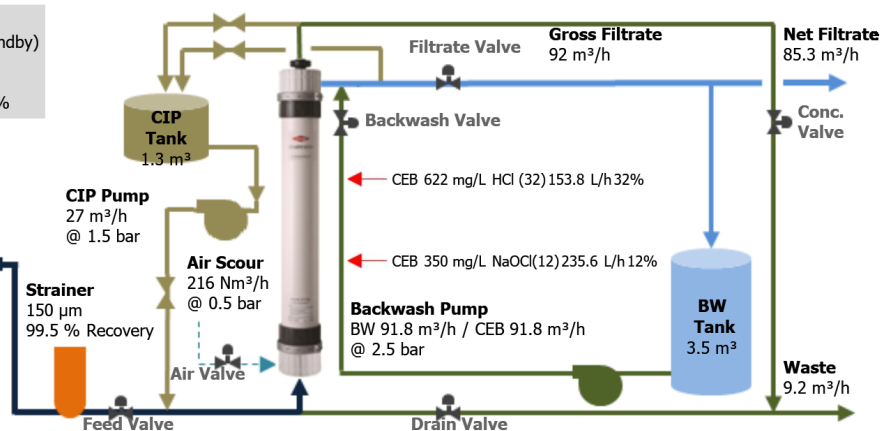
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UF Detailed Report

Module: IntegraTec™ XP 51
Total UF Units: 3 (2 + 1 standby)
UF Modules: 3 x 18 = 54
Operating Flux: 51 LMH
UF System Recovery: 90.2%

Feed Water
Average Feed Flow: 95 m³/h
Type: Surface Water
TSS: 0.0 mg/L
TOC: 0.0 mg/L
Turbidity: 0.0NTU

Feed Pump
Max 139.4 m³/h
@ 2.2 bar



UF System Overview

Module Type	Ultrafiltration IntegraTec™ XP 51		
# Units	Online = 2	Standby = 1	Redundant = 0
# Modules	Per Unit = 18	Total = 54	
System Flow Rate (m³/h)	Gross Feed = 95.0	Net Product = 85.3	
Unit Flow Rate (m³/h)	Gross Feed = 47.5	Net Product = 42.7	
UF System Recovery (%)	90.24		
TMP (bar)	0.32 @ 10.0 °C		
Utility Water	Forward Flush: Pretreated water	Backwash: UF filtrate water	
	CEB Water Source: UF filtrate water	CIP Water Source: UF filtrate water	

UF Operating Conditions

	Duration	Interval	Flux/Flow
Filtration:	30	33.2 min	-
Instantaneous			
2 Online Units			51 LMH
3 Total Units			34 LMH
Average			33 LMH
Net			31 LMH
Backwash	3.2 min	33.2 min	100 LMH
Acid CEB	15.9 min	24 h	100 LMH
Alkali CEB	15.9 min	12 h	100 LMH
CIP	311.4 min	30 d	1.50 m³/h
Membrane Integrity Testing	12.0 min	24 h	-

UF Water Quality

Stream Name	Stream 1	
Water Type	Surface Water (10.0 - 40.0 °C)	
	Feed	Expected UF Product Water Quality
Temperature (°C)	25.0	25.0
TDS (mg/L)	680	680
pH	7.0	7.0

UF Configuration Options

Standby Option:	Constant filtrate flow, variable plant module flux
Storage Tank Option:	Storage Tank sized for BW demand only
Forward Flush Water Source:	Pretreated water
BW Water Source:	UF filtrate water
CEB Water Source:	UF filtrate water
CIP Water Source:	UF filtrate water

UF System Size and Module Details

Units		Module Details		
		Name: Ultrafiltration IntegraTec™ XP 51		
Online Units	2	Membrane Area	51 m ²	549 ft ²
Standby Units	1	Length	1.860 m	73.2 in
Redundant Units	0	Diameter	0.225 m	8.9 in
Total Units	3	Weight (empty)	48 kg	106 lb
Max Offline Units	1	Weight (water filled)	83 kg	183 lb
Modules/Unit	18	Water Volume	35.0 L	9.2 gal
Total Modules	54			

UF Flow Details

Stream		Maximum Flow ^a	Average Flow
Feed (Gross)	(m ³ /h)	139.4	95.0
Feed Water Used for	(m ³ /h)		
Pretreatment	(m ³ /h)		0.5
Forward Flush & Process Streams	(m ³ /h)		2.5
Feed (Net)	(m ³ /h)	92.8	92.0
Filtrate (Gross)	(m ³ /h)	92.8	92.0
Filtrate Used for Cleaning	(m ³ /h)		6.7
Filtrate (Net)	(m ³ /h)		85.3
Air	(N m ³ /h)	216.0	5.1
Backwash (BW)	(m ³ /h)	91.8	6.1
Forward Flush Flowrate	(m ³ /h)	45.9	2.5
CEB 1 (Acid)	(m ³ /h)	91.8	0.2
CEB 2 (Alkali and/or Oxidant)	(m ³ /h)	91.8	0.4
CIP Recycle	(m ³ /h)	27.0	0.0
CEB HCl (32%) Metering Pump	(L/h)	153.8	
CEB NaOCl(12%) Metering Pump	(L/h)	235.6	
CIP HCl (32%) Metering Pump	(L/h)	45.2	
CIP Citric Acid(100%) Metering Pump	(L/h)	324.3	
CIP NaOH (50%) Metering Pump	(L/h)	33.8	
CIP NaOCl(12%) Metering Pump	(L/h)	396.0	

Footnotes:

^a Maximum possible flow rate

UF Pump Hydraulics and Electrical Cost

Pump	Peak Flowrate (m ³ /h)	Average Pressure (bar)	Mechanical Power (kW)	Electrical Power (kW)	Energy (kWh/d)	Cost (\$/d)
Feed	139.36	0.31	0.83	1.13	27.10	2.44
Backwash	91.80	0.73	1.85	2.51	4.00	0.36
CEB	91.80	0.73	1.85	2.51	0.40	0.04
HCl (32%) Metering Pump	0.15		0.00	0.00		
NaOCl(12%) Metering Pump	0.24		0.01	0.01		
CIP	27.00	1.50	1.12	1.53	0.22	0.02
HCl (32%) Metering Pump	0.05		0.00	0.00		
Citric Acid(100%) Metering Pump	0.32		0.02	0.03		
NaOH (50%) Metering Pump	0.03		0.00	0.00		
NaOCl(12%) Metering Pump	0.40		0.02	0.03		
CIP Solution Heating				87.51	2.10	0.19
Air Compressor	216.00	0.50	2.61	5.68	3.21	0.29
Electrical Valves				0.00	0.00	0.00
PLC and Instrumentation				0.30	7.20	0.65
Total Electrical Cost					44.23	3.98

UF Pressure Ratings

Process	T (°C)	TMP ^a (bar)	Fouling Max ΔP (bar)	Piping ΔP (bar)	Filtrate Pres. (bar)	Feed Pres. ^b (bar)	Pres. Rating (bar)	OK? ^c
Filtration								
Minimum Temp.	10.0	0.32	0.00	0.15	0.00	0.46	6.25	✓
Design Temp.	25.0	0.21	0.00	0.10	0.00	0.31	6.00	✓
Maximum Temp.	40.0	0.16	0.00	0.07	0.00	0.23	4.75	✓
BW	25.0	0.43	0.00	0.30		0.73	6.00	✓
CEB	25.0	0.43	0.00	0.30		0.73	6.00	✓
CIP	35.0			1.50		1.50	5.25	✓

Footnotes:

^a At actual, average flux

^b Sum of TMP, fouling ΔP, piping ΔP and filtrate pressure. Does not include pressure drop at the strainer. Pressure drops are based on user inputs. Default values should not be used for pump sizing

^c Comparison of Feed Pressure to Pressure Rating—a conservative comparison due to piping losses between the feed pump and module inlet.

UF Storage Tanks

Name	Bulk Conc. (%)	Minimum Recommended Volume (m ³)
Water ^a		3.5
CIP Tank		1.26
Chemical Storage ^b		
Citric Acid (C ₆ H ₈ O ₇)	100%	0.033
Hydrochloric Acid (HCl)	32%	0.246
Sodium Hypochlorite (NaOCl)	12%	0.79
Sodium Hydroxide (NaOH)	50%	0.003

Footnotes:

^a Storage Tank sized to maintain constant net filtrate flow

^b The minimum recommended volume for chemical storage tanks is sized for 30 days of storage.

UF Design Warnings

None

Filtration Mode and Backwash Parameters

Normal Operation	Process Mode of Operation	Operation		Backwash (B/W)					Return to Operation ^d	Stop ^e
Operating Steps	Steps	1 Forward Flush at Start-up ^b	2 Filtration Mode	3 Air Inlet ^c	4 Drain	5 Backwash 1	6 Backwash 2	7 Forward Flush (FF)	2 Filtration Mode	- Stop
Pump and Valve Conditions	Feed Pump	o	o					o	o	
	Backwash Pump					o	o			
	Chem. Dosing Pump ^a									
	CEB Dosing Pump									
	CIP Recycle Pump									
	Feed Valve	o	o					o	o	
	Filtrate Valve		o						o	
	Conc. Valve	o		o	o	o		o		
	Backwash Inlet Valve					o	o			
	Drain Valve				o		o			
	Air Inlet Valve			o						
	Duration	~2.0-3.0 min.	30.0 min.	20 s	30 s	30 s	30 s	50 s	30.0 min.	
	Flow Rate	2.6 m ³ /h	51 LMH	12.0 N m ³ /h	By gravity	100 LMH	100 LMH	2.6 m ³ /h	51 LMH	
Remarks	1. The filtration mode follows Steps 2-3-4-5-6-7. Backwash can be repeated several times according to the fouling degree of UF membrane modules. 2. The valve opening and closing time for each process step should be considered when programming is designed. 3. "o" = valve or pump is opened or operating.									
Footnotes	^a Use of chemical dosing pump during backwash is based on feed water source and quality. Refer to DuPont UF Design Guidelines. ^b Forward flush flow rate displayed on per-module basis. ^c Use of air scour and frequency is based on feed water source and quality. Air flow rate displayed on per-module basis. ^d May need to waste a portion of permeate to remove residual chemicals, depending on design and application. ^e If taken out of operation, add preservative and close all valves. Stop should occur only after backwash.									

CEB Parameters

CEB	CEB Mode of Operation	Operation	Chemically Enhanced Backwash (CEB)										Return to Operation ^d
Operating Steps	Steps	1 Filtration Mode	2 Air Inlet ^b	3 Drain	4 Backwash 1	5 Backwash 2	6 Soak	7 Air Inlet ^b	8 Drain	9 Backwash 1	10 Backwash 2	11 Forward Flush (FF) ^c	1 Filtration Mode
Pump and Valve Conditions	Feed Pump	o										o	o
	Backwash Pump				o	o				o	o		
	Chem. Dosing Pump												
	CEB Dosing Pump ^a				o	o							
	CIP Recycle Pump												
	Feed Valve	o										o	o
	Filtrate Valve	o											o
	Conc. Valve		o	o	o			o	o	o		o	
	Backwash Inlet Valve				o	o				o	o		
	Drain Valve			o		o			o		o		
	Air Inlet Valve		o					o					
	Duration	30.0 min.	20 s	30 s	45 s	45 s	10.0 min.	20 s	30 s	30 s	30 s	50 s	30.0 min.
	Flow Rate	51 LMH	12.0 N m ³ /h	By gravity	100 LMH	100 LMH	0.0 m ³ /h	12.0 N m ³ /h	By gravity	100 LMH	100 LMH	2.6 m ³ /h	51 LMH
Remarks	1. Frequency of CEB is based on feed water source and quality. Refer to DuPont UF Design Guidelines. 2. For a CEB, follow Steps 2-3-4-5-6-7-8-9-10-11. 3. "o" = valve or pump is opened or operating.												
Footnotes	^a Chemicals and concentrations used during CEB are based on feed water source and quality. Refer to DuPont UF Design Guidelines. ^b Air flow rate displayed on per-module basis. ^c Forward flush flow rate displayed on per-module basis. ^d May need to waste a portion of permeate to remove residual chemicals, depending on design and application.												

CIP Parameters

CIP	CIP Mode of Operation	Operation	Backwash (B/W) ^b				CIP ^c				Backwash (B/W) ^b				Return to Operation ^e
	Steps	1 Filtration Mode	2 Air Inlet ^a	3 Drain	4 Backwash 1	5 Backwash 2	6 Drain	7 CIP Recycle ^d	8 Soak ^e	9 CIP Recycle ^d	10 Drain	11 Backwash 1	12 Backwash 2	13 Forward Flush (FF) ^f	1 Filtration Mode
Pump and Valve Conditions	Feed Pump	o												o	o
	Backwash Pump				o	o						o	o		
	Chem. Dosing Pump														
	CEB Dosing Pump														
	CIP Recycle Pump							o		o					
	Feed Valve	o												o	o
	Filtrate Valve	o													o
	Conc. Valve		o	o	o		o		o		o	o		o	
	Backwash Inlet Valve				o	o						o	o		
	Drain Valve			o		o	o				o		o		
	Air Inlet Valve		o												
	Duration	30.0 min.	20 s	30 s	30 s	30 s	30 s	30.0 min.	90.0 min.	30.0 min.	30 s	30 s	30 s	50 s	30.0 min.
	Flow Rate	51 LMH	12.0 N m ³ /h	By gravity	100 LMH	100 LMH	By gravity	1.5 m ³ /h	0.0 m ³ /h	1.5 m ³ /h	By gravity	100 LMH	100 LMH	2.6 m ³ /h	51 LMH
Remarks	1. Frequency of CIP is 1-3 months, adjusted according to operating conditions. 2. Start CIP with backwash sequence; complete CIP with backwash sequence. 3. CIP is done manually. 4. "o" = valve or pump is opened or operating.														
Footnotes	^a Air flow rate displayed on per-module basis. ^b This step should be repeated 2 times. ^c This step and duration is shown for a single chemical cleaning. If acid and base cleaning are both required, repeat Steps 6-13. ^d CIP recycle flow rate displayed on a per-module basis. ^e The duration of this step might be longer, up to overnight (12 hours), if the fouling is severe. ^f Forward flush flow rate displayed on per-module basis. ^g May need to waste a portion of permeate to remove residual chemicals, depending on design and application.														

UF Utility and Chemical Costs

Service Water

	Average Flowrate (m ³ /h)	Unit Cost (\$/m ³)	Hourly Cost (\$/h)	Daily Cost (\$/d)
Non-Product Feed Water	9.23	0.1400	1.29	31.01
Waste Water Disposal	9.23	0.6900	6.37	152.81
Total Service Water Cost				183.82

Electricity

Peak Power	(kW)	99.76
Energy	(kWh/d)	44.23
Electricity Unit Cost	(\$/kWh)	0.0900
Electricity Cost	(\$/d)	3.98
Specific Energy	(kWh/m ³)	0.02

Chemicals

Chemical	Unit Cost (\$/kg)	Dose (mg/L)	Volume (L/d)	Cost (\$/d)
Citric Acid (100%)	1.520		1.1	2.76
CIP		20000		
HCl (32%)	0.100		8.2	0.95
CEB1		622		
CIP		622		
NaOCl (12%)	0.330		26.4	9.90
CEB2		350		
CIP		2000		
NaOH (50%)	0.258		0.1	0.04
CIP		955		
Total Chemical Cost				13.66

Utility and Chemical Cost	(\$/d)	201.46
Specific Water Cost	(\$/m ³)	0.098

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